Area of Trapeziums, Kites and Circles

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

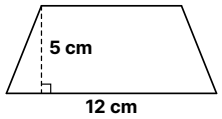
KEY WORDS area *(horahanga)* · trapezium *(tapawhā whakarara-tahi)* · parallel sides · kite *(manu aute)*

A Trapeziums ●●○

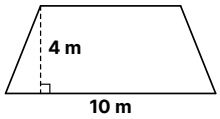
Remember: Area of a trapezium $= \frac{1}{2}(a + b) \times h$: add the two parallel sides, halve, then multiply by the height between them.

1. Find the area of each trapezium.

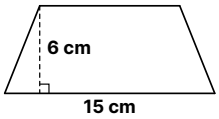
a)



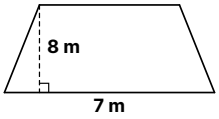
b)



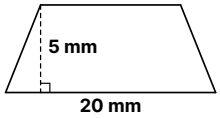
c)



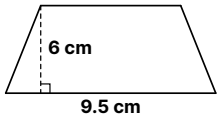
d)



e)



f)



2. Find the area of:

a) A trapezium with parallel sides of 14 cm and 10 cm, and a height of 7 cm

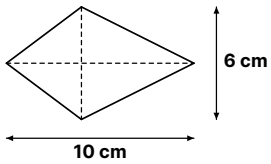
b) A trapezium with parallel sides of 2.5 m and 1.5 m, and a height of 3 m

B Kites ●●○

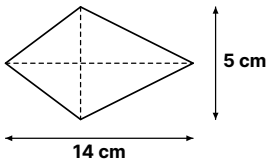
Remember: Area of a kite $= \frac{1}{2} \times$ one diagonal $\times$ the other diagonal.

3. Find the area of each kite.

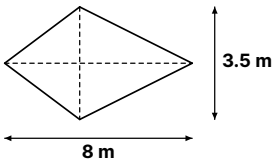
a)



b)



c)



4. Find the area of:

a) A kite with diagonals of 20 cm and 13 cm

b) A kite with diagonals of 1.2 m and 0.5 m

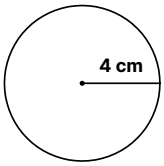
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C Circles ●●○

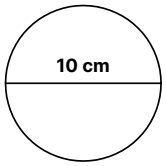
> **Remember:** Area of a circle $= \pi r^2$. If you are given the diameter, halve it first to get the radius. Round to 1 decimal place.

5. Find the area of each circle.

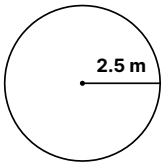
a)



b)



c)



6. Find:

a) The area of a circle with radius 7 cm

b) The area of a circle with diameter 18 mm

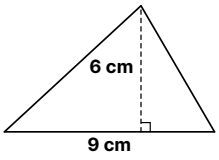
c) The area of a circle with radius 0.6 m

d) The area of a circle with diameter 5 km

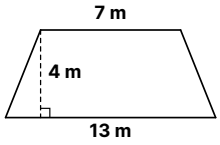
D Choosing the formula ●●●

7. Decide which formula each shape needs, then find its area. Round to 1 decimal place where needed.

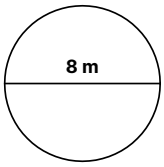
a)



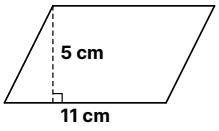
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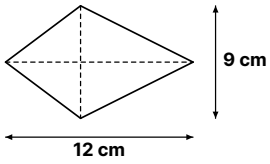
c)



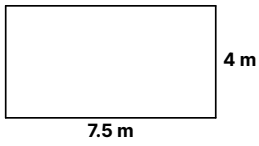
d)



e)



f)



E Problems ●●●

8. A circular rug has a diameter of 2.4 m. Find its area, to 1 decimal place.

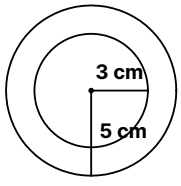
9. A garden is a trapezium. Its parallel sides are 18 m and 12 m long, and they are 10 m apart. Find the area of the garden.

10. Which has more pizza: one pizza with a diameter of 30 cm, or two pizzas each with a diameter of 20 cm? Show your working.

★ Extension

11. Find the area of a semicircle with a diameter of 12 cm, to 1 decimal place.

12. A circle with a radius of 3 cm is cut out of a circle with a radius of 5 cm. Find the area of the ring that is left, to 1 decimal place.



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Area of Trapeziums, Kites and Circles

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

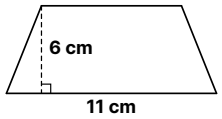
KEY WORDS area *(horahanga)* · trapezium *(tapawhā whakarara-tahi)* · parallel sides · kite *(manu aute)*

A Trapeziums ●●○

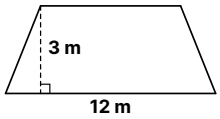
Remember: Area of a trapezium $= \frac{1}{2}(a + b) \times h$: add the two parallel sides, halve, then multiply by the height between them.

1. Find the area of each trapezium.

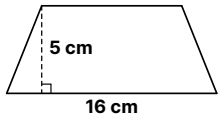
a)



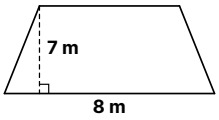
b)



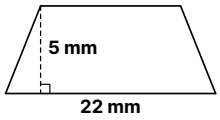
c)



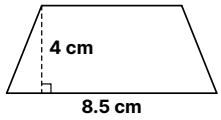
d)



e)



f)



2. Find the area of:

a) A trapezium with parallel sides of 13 cm and 9 cm, and a height of 8 cm

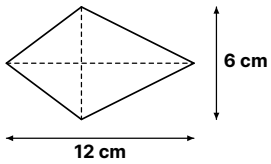
b) A trapezium with parallel sides of 3.5 m and 2.5 m, and a height of 2 m

B Kites ●●○

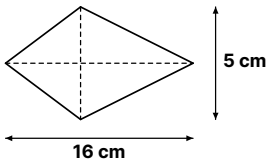
Remember: Area of a kite $= \frac{1}{2} \times$ one diagonal $\times$ the other diagonal.

3. Find the area of each kite.

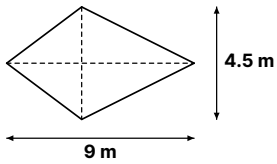
a)



b)



c)



4. Find the area of:

a) A kite with diagonals of 24 cm and 11 cm

b) A kite with diagonals of 1.6 m and 0.5 m

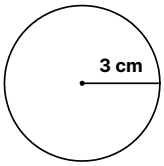
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C Circles ●●○

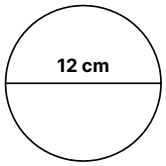
> **Remember:** Area of a circle $= \pi r^2$. If you are given the diameter, halve it first to get the radius. Round to 1 decimal place.

5. Find the area of each circle.

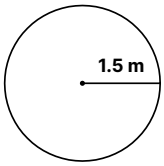
a)



b)



c)



6. Find:

a) The area of a circle with radius 9 cm

b) The area of a circle with diameter 14 mm

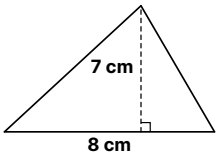
c) The area of a circle with radius 0.8 m

d) The area of a circle with diameter 3 km

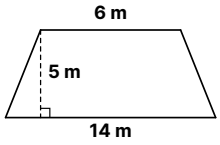
D Choosing the formula ●●●

7. Decide which formula each shape needs, then find its area. Round to 1 decimal place where needed.

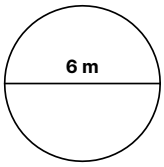
a)



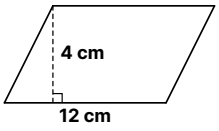
b)



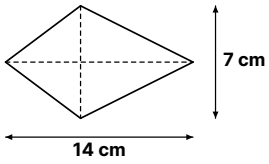
c)



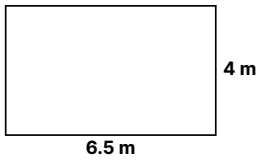
d)



e)



f)



E Problems ●●●

8. A circular rug has a diameter of 3.2 m. Find its area, to 1 decimal place.

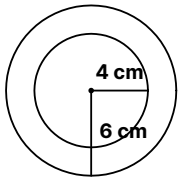
9. A garden is a trapezium. Its parallel sides are 20 m and 14 m long, and they are 8 m apart. Find the area of the garden.

10. Which has more pizza: one pizza with a diameter of 40 cm, or two pizzas each with a diameter of 28 cm? Show your working.

★ Extension

11. Find the area of a semicircle with a diameter of 16 cm, to 1 decimal place.

12. A circle with a radius of 4 cm is cut out of a circle with a radius of 6 cm. Find the area of the ring that is left, to 1 decimal place.



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